

بخشی از مهمترین رجیسترهای atmega 32

PDIP

(XCK/T0) PB0	☐	1	40	☐	PA0 (ADC0)
(T1) PB1	☐	2	39	☐	PA1 (ADC1)
(INT2/AIN0) PB2	☐	3	38	☐	PA2 (ADC2)
(OC0/AIN1) PB3	☐	4	37	☐	PA3 (ADC3)
($\overline{SS}$) PB4	☐	5	36	☐	PA4 (ADC4)
(MOSI) PB5	☐	6	35	☐	PA5 (ADC5)
(MISO) PB6	☐	7	34	☐	PA6 (ADC6)
(SCK) PB7	☐	8	33	☐	PA7 (ADC7)
$\overline{RESET}$	☐	9	32	☐	AREF
VCC	☐	10	31	☐	GND
GND	☐	11	30	☐	AVCC
XTAL2	☐	12	29	☐	PC7 (TOSC2)
XTAL1	☐	13	28	☐	PC6 (TOSC1)
(RXD) PD0	☐	14	27	☐	PC5 (TDI)
(TXD) PD1	☐	15	26	☐	PC4 (TDO)
(INT0) PD2	☐	16	25	☐	PC3 (TMS)
(INT1) PD3	☐	17	24	☐	PC2 (TCK)
(OC1B) PD4	☐	18	23	☐	PC1 (SDA)
(OC1A) PD5	☐	19	22	☐	PC0 (SCL)
(ICP1) PD6	☐	20	21	☐	PD7 (OC2)

Interrupt Vectors in ATmega32

Table 18. Reset and Interrupt Vectors

Vector No.	Program Address ⁽²⁾	Source	Interrupt Definition
1	\$000 ⁽¹⁾	RESET	External Pin, Power-on Reset, Brown-out Reset, Watchdog Reset, and JTAG AVR Reset
2	\$002	INT0	External Interrupt Request 0
3	\$004	INT1	External Interrupt Request 1
4	\$006	INT2	External Interrupt Request 2
5	\$008	TIMER2 COMP	Timer/Counter2 Compare Match
6	\$00A	TIMER2 OVF	Timer/Counter2 Overflow
7	\$00C	TIMER1 CAPT	Timer/Counter1 Capture Event
8	\$00E	TIMER1 COMPA	Timer/Counter1 Compare Match A
9	\$010	TIMER1 COMPB	Timer/Counter1 Compare Match B
10	\$012	TIMER1 OVF	Timer/Counter1 Overflow
11	\$014	TIMER0 COMP	Timer/Counter0 Compare Match
12	\$016	TIMER0 OVF	Timer/Counter0 Overflow
13	\$018	SPI, STC	Serial Transfer Complete
14	\$01A	USART, RXC	USART, Rx Complete
15	\$01C	USART, UDRE	USART Data Register Empty
16	\$01E	USART, TXC	USART, Tx Complete
17	\$020	ADC	ADC Conversion Complete
18	\$022	EE_RDY	EEPROM Ready
19	\$024	ANA_COMP	Analog Comparator
20	\$026	TWI	Two-wire Serial Interface
21	\$028	SPM_RDY	Store Program Memory Ready

- GIFR: ثبت پرچم وقفه‌ها

General Interrupt Flag Register – GIFR

Bit	7	6	5	4	3	2	1	0	
	INTF1	INTF0	INTF2	–	–	–	–	–	GIFR
Read/Write	R/W	R/W	R/W	R	R	R	R	R	
Initial Value	0	0	0	0	0	0	0	0	

- MCUCR: ثبت کنترل وقفه‌ها و حالت خواب

**MCU Control Register
– MCUCR**

The MCU Control Register contains control bits for interrupt sense control and general MCU functions.

Bit	7	6	5	4	3	2	1	0	
	SE	SM2	SM1	SM0	ISC11	ISC10	ISC01	ISC00	MCUCR
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

• **Bit 3, 2 – ISC11, ISC10: Interrupt Sense Control 1 Bit 1 and Bit 0**

The External Interrupt 1 is activated by the external pin INT1 if the SREG I-bit and the corresponding interrupt mask in the GICR are set. The level and edges on the external INT1 pin that activate the interrupt are defined in [Table 34](#). The value on the INT1 pin is sampled before detecting edges. If edge or toggle interrupt is selected, pulses that last longer than one clock period will generate an interrupt. Shorter pulses are not guaranteed to generate an interrupt. If low level interrupt is selected, the low level must be held until the completion of the currently executing instruction to generate an interrupt.

Table 34. Interrupt 1 Sense Control

ISC11	ISC10	Description
0	0	The low level of INT1 generates an interrupt request.
0	1	Any logical change on INT1 generates an interrupt request.
1	0	The falling edge of INT1 generates an interrupt request.
1	1	The rising edge of INT1 generates an interrupt request.

3. ADC (Analog to Digital Converter Registers):

- ADMUX: (ADC) ثبت ورودی مولتیپلکسر آنالوگ به دیجیتال

ADC Multiplexer
Selection Register –
ADMUX

Bit	7	6	5	4	3	2	1	0	
	REFS1	REFS0	ADLAR	MUX4	MUX3	MUX2	MUX1	MUX0	ADMUX
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 83. Voltage Reference Selections for ADC

REFS1	REFS0	Voltage Reference Selection
0	0	AREF, Internal Vref turned off
0	1	AVCC with external capacitor at AREF pin
1	0	Reserved
1	1	Internal 2.56V Voltage Reference with external capacitor at AREF pin

Table 84. Input Channel and Gain Selections

MUX4..0	Single Ended Input	Positive Differential Input	Negative Differential Input	Gain
00000	ADC0	N/A		
00001	ADC1			
00010	ADC2			
00011	ADC3			
00100	ADC4			
00101	ADC5			
00110	ADC6			
00111	ADC7			
01000	N/A	ADC0	ADC0	10x
01001		ADC1	ADC0	
01010		ADC0	ADC0	200x
01011		ADC1	ADC0	
01100		ADC2	ADC2	10x
01101		ADC3	ADC2	
01110		ADC2	ADC2	200x
01111		ADC3	ADC2	
10000		ADC0	ADC1	1x
10001		ADC1	ADC1	
10010		ADC2	ADC1	
10011		ADC3	ADC1	
10100		ADC4	ADC1	
10101		ADC5	ADC1	
10110		ADC6	ADC1	
10111		ADC7	ADC1	
11000		ADC0	ADC2	
11001		ADC1	ADC2	
11010		ADC2	ADC2	
11011		ADC3	ADC2	
11100		ADC4	ADC2	

ADC ثبت کنترول: ADCSRA -

Bit	7	6	5	4	3	2	1	0
	ADEN	ADSC	ADATE	ADIF	ADIE	ADPS2	ADPS1	ADPS0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

ADPS2	ADPS1	ADPS0	Division Factor
0	0	0	2
0	0	1	2
0	1	0	4
0	1	1	8
1	0	0	16
1	0	1	32
1	1	0	64
1	1	1	128

$$ADLAR = 0$$

Bit	15	14	13	12	11	10	9	8	
	-	-	-	-	-	-	ADC9	ADC8	ADCH
	ADC7	ADC6	ADC5	ADC4	ADC3	ADC2	ADC1	ADC0	ADCL
	7	6	5	4	3	2	1	0	
Read/Write	R	R	R	R	R	R	R	R	
	R	R	R	R	R	R	R	R	
Initial Value	0	0	0	0	0	0	0	0	
	0	0	0	0	0	0	0	0	

[illegible]

5. تایمر / کانتر (Timer/Counter Registers):

- TCCR0: ثبت کنترل تایمر/شمارنده صفر

Timer/Counter Control
Register – TCCR0

Bit	7	6	5	4	3	2	1	0	
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00	TCCR0
Read/Write	W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 38. Waveform Generation Mode Bit Description⁽¹⁾

Mode	WGM01 (CTC0)	WGM00 (PWM0)	Timer/Counter Mode of Operation	TOP	Update of OCR0	TOV0 Flag Set-on
0	0	0	Normal	0xFF	Immediate	MAX
1	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	1	0	CTC	OCR0	Immediate	MAX
3	1	1	Fast PWM	0xFF	BOTTOM	MAX

Note: 1. The CTC0 and PWM0 bit definition names are now obsolete. Use the WGM01:0 definitions. However, the functionality and location of these bits are compatible with previous versions of the timer.

Table 39. Compare Output Mode, non-PWM Mode

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected.
0	1	Toggle OC0 on compare match
1	0	Clear OC0 on compare match
1	1	Set OC0 on compare match

Table 40 shows the COM01:0 bit functionality when the WGM01:0 bits are set to fast PWM mode.

Table 40. Compare Output Mode, Fast PWM Mode⁽¹⁾

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected.
0	1	Reserved
1	0	Clear OC0 on compare match, set OC0 at BOTTOM, (non-inverting mode)
1	1	Set OC0 on compare match, clear OC0 at BOTTOM, (inverting mode)

Note: 1. A special case occurs when OCR0 equals TOP and COM01 is set. In this case, the compare match is ignored, but the set or clear is done at BOTTOM. See [“Fast PWM Mode” on page 75](#) for more details.

Table 41. Compare Output Mode, Phase Correct PWM Mode⁽¹⁾

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected.
0	1	Reserved
1	0	Clear OC0 on compare match when up-counting. Set OC0 on compare match when downcounting.
1	1	Set OC0 on compare match when up-counting. Clear OC0 on compare match when downcounting.

Note: 1. A special case occurs when OCR0 equals TOP and COM01 is set. In this case, the compare match is ignored, but the set or clear is done at TOP. See [“Phase Correct PWM Mode” on page 76](#) for more details.

- **Bit 2:0 – CS02:0: Clock Select**

The three Clock Select bits select the clock source to be used by the Timer/Counter.

Table 42. Clock Select Bit Description

CS02	CS01	CS00	Description
0	0	0	No clock source (Timer/Counter stopped).
0	0	1	clk _{I/O} /(No prescaling)
0	1	0	clk _{I/O} /8 (From prescaler)
0	1	1	clk _{I/O} /64 (From prescaler)
1	0	0	clk _{I/O} /256 (From prescaler)
1	0	1	clk _{I/O} /1024 (From prescaler)
1	1	0	External clock source on T0 pin. Clock on falling edge.
1	1	1	External clock source on T0 pin. Clock on rising edge.

- **OCR0:** ثبت مقادیر تنظیم شده برای PWM در تایمر/شمارنده صفر

**Output Compare
Register – OCR0**

Bit	7	6	5	4	3	2	1	0	
	OCR0[7:0]								OCR0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **TCCR1A و TCCR1B:** ثبت کنترل تایمر/شمارنده یک

**Timer/Counter1
Control Register A –
TCCR1A**

Bit	7	6	5	4	3	2	1	0	
	COM1A1	COM1A0	COM1B1	COM1B0	FOC1A	FOC1B	WGM11	WGM10	TCCR1A
Read/Write	R/W	R/W	R/W	R/W	W	W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 44. Compare Output Mode, non-PWM

COM1A1/COM1B1	COM1A0/COM1B0	Description
0	0	Normal port operation, OC1A/OC1B disconnected.
0	1	Toggle OC1A/OC1B on compare match
1	0	Clear OC1A/OC1B on compare match (Set output to low level)
1	1	Set OC1A/OC1B on compare match (Set output to high level)

Table 45. Compare Output Mode, Fast PWM⁽¹⁾

COM1A1/COM1B1	COM1A0/COM1B0	Description
0	0	Normal port operation, OC1A/OC1B disconnected.
0	1	WGM13:0 = 15: Toggle OC1A on Compare Match, OC1B disconnected (normal port operation). For all other WGM13:0 settings, normal port operation, OC1A/OC1B disconnected.
1	0	Clear OC1A/OC1B on compare match, set OC1A/OC1B at BOTTOM, (non-inverting mode)
1	1	Set OC1A/OC1B on compare match, clear OC1A/OC1B at BOTTOM, (inverting mode)

Note: 1. A special case occurs when OCR1A/OCR1B equals TOP and COM1A1/COM1B1 is set. In this case the compare match is ignored, but the set or clear is done at BOTTOM. See [“Fast PWM Mode” on page 99.](#) for more details.

Table 46. Compare Output Mode, Phase Correct and Phase and Frequency Correct PWM ⁽¹⁾

COM1A1/COM1B1	COM1A0/COM1B0	Description
0	0	Normal port operation, OC1A/OC1B disconnected.
0	1	WGM13:0 = 9 or 14: Toggle OC1A on Compare Match, OC1B disconnected (normal port operation). For all other WGM13:0 settings, normal port operation, OC1A/OC1B disconnected.
1	0	Clear OC1A/OC1B on compare match when up-counting. Set OC1A/OC1B on compare match when downcounting.
1	1	Set OC1A/OC1B on compare match when up-counting. Clear OC1A/OC1B on compare match when downcounting.

Note: 1. A special case occurs when OCR1A/OCR1B equals TOP and COM1A1/COM1B1 is set. See [“Phase Correct PWM Mode” on page 101.](#) for more details.

Table 47. Waveform Generation Mode Bit Description⁽¹⁾

Mode	WGM13	WGM12 (CTC1)	WGM11 (PWM11)	WGM10 (PWM10)	Timer/Counter Mode of Operation	TOP	Update of OCR1x	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	BOTTOM	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	BOTTOM	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	BOTTOM	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	Reserved	–	–	–
14	1	1	1	0	Fast PWM	ICR1	BOTTOM	TOP
15	1	1	1	1	Fast PWM	OCR1A	BOTTOM	TOP

Note: 1. The CTC1 and PWM11:0 bit definition names are obsolete. Use the WGM12:0 definitions. However, the functionality and location of these bits are compatible with previous versions of the timer.

Timer/Counter1 Control Register B – TCCR1B

Bit	7	6	5	4	3	2	1	0	
	ICNC1	ICES1	–	WGM13	WGM12	CS12	CS11	CS10	TCCR1B
Read/Write	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 48. Clock Select Bit Description

CS12	CS11	CS10	Description
0	0	0	No clock source (Timer/Counter stopped).
0	0	1	$\text{clk}_{\text{IO}}/1$ (No prescaling)
0	1	0	$\text{clk}_{\text{IO}}/8$ (From prescaler)
0	1	1	$\text{clk}_{\text{IO}}/64$ (From prescaler)
1	0	0	$\text{clk}_{\text{IO}}/256$ (From prescaler)
1	0	1	$\text{clk}_{\text{IO}}/1024$ (From prescaler)
1	1	0	External clock source on T1 pin. Clock on falling edge.
1	1	1	External clock source on T1 pin. Clock on rising edge.

ثبت مقادیر تنظیم شده برای PWM در تایمر/شمارنده یک: OCR1A و OCR1B

Output Compare

Register 1 A –

OCR1AH and OCR1AL

Bit	7	6	5	4	3	2	1	0	
	OCR1A[15:8]								OCR1AH
	OCR1A[7:0]								OCR1AL
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Output Compare

Register 1 B –

OCR1BH and OCR1BL

Bit	7	6	5	4	3	2	1	0	
	OCR1B[15:8]								OCR1BH
	OCR1B[7:0]								OCR1BL
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

ثبت مقدار شمارش تایمر/شمارنده صفر: TCNT0

Timer/Counter

Register – TCNT0

Bit	7	6	5	4	3	2	1	0	
	TCNT0[7:0]								TCNT0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

ثبت مقدار شمارش تایمر/شمارنده یک: TCNT1

Timer/Counter1 –

TCNT1H and TCNT1L

Bit	7	6	5	4	3	2	1	0	
	TCNT1[15:8]								TCNT1H
	TCNT1[7:0]								TCNT1L
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

ثبت کنترل تایمر/شمارنده دو: TCCR2

Timer/Counter Control Register – TCCR2

Bit	7	6	5	4	3	2	1	0	
	FOC2	WGM20	COM21	COM20	WGM21	CS22	CS21	CS20	TCCR2
Read/Write	W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Table 50. Waveform Generation Mode Bit Description⁽¹⁾

Mode	WGM21 (CTC2)	WGM20 (PWM2)	Timer/Counter Mode of Operation	TOP	Update of OCR2	TOV2 Flag Set on
0	0	0	Normal	0xFF	Immediate	MAX
1	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	1	0	CTC	OCR2	Immediate	MAX
3	1	1	Fast PWM	0xFF	BOTTOM	MAX

Note: 1. The CTC2 and PWM2 bit definition names are now obsolete. Use the WGM21:0 definitions. However, the functionality and location of these bits are compatible with previous versions of the timer.

Table 51. Compare Output Mode, non-PWM Mode

COM21	COM20	Description
0	0	Normal port operation, OC2 disconnected.
0	1	Toggle OC2 on compare match
1	0	Clear OC2 on compare match
1	1	Set OC2 on compare match

Table 52 shows the COM21:0 bit functionality when the WGM21:0 bits are set to fast PWM mode.

Table 53. Compare Output Mode, Phase Correct PWM Mode⁽¹⁾

COM21	COM20	Description
0	0	Normal port operation, OC2 disconnected.
0	1	Reserved
1	0	Clear OC2 on compare match when up-counting. Set OC2 on compare match when downcounting.
1	1	Set OC2 on compare match when up-counting. Clear OC2 on compare match when downcounting.

Note: 1. A special case occurs when OCR2 equals TOP and COM21 is set. In this case, the compare match is ignored, but the set or clear is done at TOP. See “Phase Correct PWM Mode” on page 121 for more details.

CS22	CS21	CS20	Description
0	0	0	No clock source (Timer/Counter stopped).
0	0	1	clk _{T2S} /(No prescaling)
0	1	0	clk _{T2S} /8 (From prescaler)
0	1	1	clk _{T2S} /32 (From prescaler)
1	0	0	clk _{T2S} /64 (From prescaler)
1	0	1	clk _{T2S} /128 (From prescaler)
1	1	0	clk _{T2S} /256 (From prescaler)
1	1	1	clk _{T2S} /1024 (From prescaler)

Bit	7	6	5	4	3	2	1	0
	OCR2[7:0]							OCR2
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

6. SPI (Serial Peripheral Interface Registers):

(SPI) ثبت کنترول رابطہ سریال یریفرا ل - SPCR:

SPI Control Register – SPCR

[illegible]

Table 56. CPOL Functionality

CPOL	Leading Edge	Trailing Edge
0	Rising	Falling
1	Falling	Rising

Table 57. CPHA Functionality

CPHA	Leading Edge	Trailing Edge
0	Sample	Setup
1	Setup	Sample

Table 58. Relationship Between SCK and the Oscillator Frequency

SPI2X	SPR1	SPR0	SCK Frequency
0	0	0	$f_{osc}/4$
0	0	1	$f_{osc}/16$
0	1	0	$f_{osc}/64$
0	1	1	$f_{osc}/128$
1	0	0	$f_{osc}/2$
1	0	1	$f_{osc}/8$
1	1	0	$f_{osc}/32$
1	1	1	$f_{osc}/64$

(SPI) ثبت وضعیت رابط سریال یریفرا ل - SPSR:

SPI Status Register – SPSR

[illegible]

(SPI) ثبت داده‌های ارسالی و دریافتی رابط سریال پرفرال: SPDR -

SPI Data Register – SPDR

Bit	7	6	5	4	3	2	1	0	
	MSB							LSB	SPDR
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	X	X	X	X	X	X	X	X	Undefined

7. USART (Universal Synchronous/Asynchronous Receiver/Transmitter) Registers

- UCSRnA: ثبت کنترل عملکرد USART

USART Control and Status Register A – UCSRA

Bit	7	6	5	4	3	2	1	0	
	RXC	TXC	UDRE	FE	DOR	PE	U2X	MPCM	UCSRA
Read/Write	R	R/W	R	R	R	R	R/W	R/W	
Initial Value	0	0	1	0	0	0	0	0	

- UCSRnB: ثبت کنترل روشن/خاموش بودن و تنظیمات دیگر

USART Control and Status Register B – UCSRB

Bit	7	6	5	4	3	2	1	0	
	RXCIE	TXCIE	UDRIE	RXEN	TXEN	UCSZ2	RXB8	TXB8	UCSRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- UCSRnC: ثبت کنترل حالت و تنظیمات دیگر

USART Control and Status Register C – UCSRC

Bit	7	6	5	4	3	2	1	0	
	URSEL	UMSEL	UPM1	UPM0	USBS	UCSZ1	UCSZ0	UCPOL	UCSRC
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	1	0	0	0	0	1	1	0	

Table 63. UMSEL Bit Settings

UMSEL	Mode
0	Asynchronous Operation
1	Synchronous Operation

Table 64. UPM Bits Settings

UPM1	UPM0	Parity Mode
0	0	Disabled
0	1	Reserved
1	0	Enabled, Even Parity
1	1	Enabled, Odd Parity

Table 65. USBS Bit Settings

USBS	Stop Bit(s)
0	1-bit
1	2-bit

Table 66. UCSZ Bits Settings

UCSZ2	UCSZ1	UCSZ0	Character Size
0	0	0	5-bit
0	0	1	6-bit
0	1	0	7-bit
0	1	1	8-bit
1	0	0	Reserved
1	0	1	Reserved
1	1	0	Reserved
1	1	1	9-bit

Table 67. UCPOL Bit Settings

UCPOL	Transmitted Data Changed (Output of TxD Pin)	Received Data Sampled (Input on RxD Pin)
0	Rising XCK Edge	Falling XCK Edge
1	Falling XCK Edge	Rising XCK Edge

USART ثبت تنظیمات سرعت ارتباط - **UBRRn**

**USART Baud Rate
Registers – UBRRL
and UBRRH**

Bit	15	14	13	12	11	10	9	8																	
	<table><tr><td>URSEL</td><td>–</td><td>–</td><td>–</td><td colspan="4">UBRR[11:8]</td></tr><tr><td colspan="8">UBRR[7:0]</td></tr></table>								URSEL	–	–	–	UBRR[11:8]				UBRR[7:0]								UBRRH UBRRL
URSEL	–	–	–	UBRR[11:8]																					
UBRR[7:0]																									
	7	6	5	4	3	2	1	0																	
Read/Write	R/W	R	R	R	R/W	R/W	R/W	R/W																	
	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W																	
Initial Value	0	0	0	0	0	0	0	0																	
	0	0	0	0	0	0	0	0																	

Table 68. Examples of UBRR Settings for Commonly Used Oscillator Frequencies

Baud Rate (bps)	$f_{osc} = 1.0000\text{MHz}$				$f_{osc} = 1.8432\text{MHz}$				$f_{osc} = 2.0000\text{MHz}$			
	U2X = 0		U2X = 1		U2X = 0		U2X = 1		U2X = 0		U2X = 1	
	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error
2400	25	0.2%	51	0.2%	47	0.0%	95	0.0%	51	0.2%	103	0.2%
4800	12	0.2%	25	0.2%	23	0.0%	47	0.0%	25	0.2%	51	0.2%
9600	6	-7.0%	12	0.2%	11	0.0%	23	0.0%	12	0.2%	25	0.2%
14.4k	3	8.5%	8	-3.5%	7	0.0%	15	0.0%	8	-3.5%	16	2.1%
19.2k	2	8.5%	6	-7.0%	5	0.0%	11	0.0%	6	-7.0%	12	0.2%
28.8k	1	8.5%	3	8.5%	3	0.0%	7	0.0%	3	8.5%	8	-3.5%
38.4k	1	-18.6%	2	8.5%	2	0.0%	5	0.0%	2	8.5%	6	-7.0%
57.6k	0	8.5%	1	8.5%	1	0.0%	3	0.0%	1	8.5%	3	8.5%
76.8k	–	–	1	-18.6%	1	-25.0%	2	0.0%	1	-18.6%	2	8.5%
115.2k	–	–	0	8.5%	0	0.0%	1	0.0%	0	8.5%	1	8.5%
230.4k	–	–	–	–	–	–	0	0.0%	–	–	–	–
250k	–	–	–	–	–	–	–	–	–	–	0	0.0%
Max ⁽¹⁾	62.5 Kbps		125 Kbps		115.2 Kbps		230.4 Kbps		125 Kbps		250 Kbps	

1. UBRR = 0, Error = 0.0%

Table 69. Examples of UBRR Settings for Commonly Used Oscillator Frequencies (Continued)

Baud Rate (bps)	$f_{osc} = 3.6864\text{MHz}$				$f_{osc} = 4.0000\text{MHz}$				$f_{osc} = 7.3728\text{MHz}$			
	U2X = 0		U2X = 1		U2X = 0		U2X = 1		U2X = 0		U2X = 1	
	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error
2400	95	0.0%	191	0.0%	103	0.2%	207	0.2%	191	0.0%	383	0.0%
4800	47	0.0%	95	0.0%	51	0.2%	103	0.2%	95	0.0%	191	0.0%
9600	23	0.0%	47	0.0%	25	0.2%	51	0.2%	47	0.0%	95	0.0%
14.4k	15	0.0%	31	0.0%	16	2.1%	34	-0.8%	31	0.0%	63	0.0%
19.2k	11	0.0%	23	0.0%	12	0.2%	25	0.2%	23	0.0%	47	0.0%
28.8k	7	0.0%	15	0.0%	8	-3.5%	16	2.1%	15	0.0%	31	0.0%
38.4k	5	0.0%	11	0.0%	6	-7.0%	12	0.2%	11	0.0%	23	0.0%
57.6k	3	0.0%	7	0.0%	3	8.5%	8	-3.5%	7	0.0%	15	0.0%
76.8k	2	0.0%	5	0.0%	2	8.5%	6	-7.0%	5	0.0%	11	0.0%
115.2k	1	0.0%	3	0.0%	1	8.5%	3	8.5%	3	0.0%	7	0.0%
230.4k	0	0.0%	1	0.0%	0	8.5%	1	8.5%	1	0.0%	3	0.0%
250k	0	-7.8%	1	-7.8%	0	0.0%	1	0.0%	1	-7.8%	3	-7.8%
0.5M	—	—	0	-7.8%	—	—	0	0.0%	0	-7.8%	1	-7.8%
1M	—	—	—	—	—	—	—	—	—	—	0	-7.8%
Max ⁽¹⁾	230.4Kbps		460.8Kbps		250Kbps		0.5Mbps		460.8Kbps		921.6Kbps	

1. UBRR = 0, Error = 0.0%

Table 70. Examples of UBRR Settings for Commonly Used Oscillator Frequencies (Continued)

Baud Rate (bps)	$f_{osc} = 8.0000\text{MHz}$				$f_{osc} = 11.0592\text{MHz}$				$f_{osc} = 14.7456\text{MHz}$			
	U2X = 0		U2X = 1		U2X = 0		U2X = 1		U2X = 0		U2X = 1	
	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error	UBRR	Error
2400	207	0.2%	416	-0.1%	287	0.0%	575	0.0%	383	0.0%	767	0.0%
4800	103	0.2%	207	0.2%	143	0.0%	287	0.0%	191	0.0%	383	0.0%
9600	51	0.2%	103	0.2%	71	0.0%	143	0.0%	95	0.0%	191	0.0%
14.4k	34	-0.8%	68	0.6%	47	0.0%	95	0.0%	63	0.0%	127	0.0%
19.2k	25	0.2%	51	0.2%	35	0.0%	71	0.0%	47	0.0%	95	0.0%
28.8k	16	2.1%	34	-0.8%	23	0.0%	47	0.0%	31	0.0%	63	0.0%
38.4k	12	0.2%	25	0.2%	17	0.0%	35	0.0%	23	0.0%	47	0.0%
57.6k	8	-3.5%	16	2.1%	11	0.0%	23	0.0%	15	0.0%	31	0.0%
76.8k	6	-7.0%	12	0.2%	8	0.0%	17	0.0%	11	0.0%	23	0.0%
115.2k	3	8.5%	8	-3.5%	5	0.0%	11	0.0%	7	0.0%	15	0.0%
230.4k	1	8.5%	3	8.5%	2	0.0%	5	0.0%	3	0.0%	7	0.0%
250k	1	0.0%	3	0.0%	2	-7.8%	5	-7.8%	3	-7.8%	6	5.3%
0.5M	0	0.0%	1	0.0%	—	—	2	-7.8%	1	-7.8%	3	-7.8%
1M	—	—	0	0.0%	—	—	—	—	0	-7.8%	1	-7.8%
Max ⁽¹⁾	0.5Mbps		1Mbps		691.2Kbps		1.3824Mbps		921.6Kbps		1.8432Mbps	

USART ثبت داده‌های دریافتی و ارسالی در: UDRn -

